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CSA-1603 COS AT-2

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Slot - C

1. Apriori Algorithm

Input :

Transaction database D

Minimum support Min-support

Output :

Frequent itemset

Begin :

1. Generate candidate 1-itemset C1 from D.
2. Count the support of every item in C1.
3. Remove itemset whose support is less than Min-support.
4. Store the remaining itemset as L1.
5. SET K=2.
6. While L(K-1) is not empty DO

Generate candidate k-itemset Ck
by joining itemset in L(k-1)

For each candidate c in Ck Do
Count the no of transaction contain c
calculate support (c).

If support (c) < Min-support then
prune c.
end If
end For

Store remaining candidates as Lk.

set k = k+1

end while.

7, Return all frequent itemsets

End.

For the given dataset, if Min support = 60%.

{Mobile} → 4/5 = 80%.

{Earphone} → 3/5 = 60%.

{charger} → 3/5 = 60%.

{Mobile, earphones} → 3/5 = 60%.

{Mobile, charger} → 2/5 = 40% → pruned

{Earphones, charger} → 2/5 = 40% → pruned

2. FP- Growth Algorithm

Input :

Transaction database D

Minimum support min_support

Output :

Frequent itemsets

Begin :

1, scan D & calculate support of every item

2, Remove item whose support is less than min_support

3, sort remaining item in descending order of support.

4, Create an empty FP-Tree & header table.

5, For each transaction T in D do

Remove infrequent item from T.

sort the remaining items according to

the global frequency order

Insert the ordered transaction into the FP-tree

If the node already exist then

Increase its count

Else

Create a new node

End if

6, Create links between node contains the same item

7, start from the least frequent item.

8, construct its conditional pattern base.

9, construct the conditional FP-tree.

10, Recursively mine the conditional FP-tree.

11, combine the discovered patterns with the suffix item.

12, Repeat until all items are processed.

13, Return all frequent itemset

END.

For the given records;

python → 4

Machine learning → 4

Data science → 3

3. Algorithm Rule Generation

Input :

Frequent itemset F
Support counts
Minimum confidence min-confidence

Output :

strong association rules

Begin :

1, For each frequent itemset I in F DO

If size(I) >= 2 Then

Generate all non-empty proper subset x of I.

For each subset x DO

$y = I - x$

confidence ($x \to y$)

$= support(x \cup y) / support(x)$

If confidence $(X \to Y) >=$ Min_confidence then
Add $X \to Y$ to the strong rules.
End if
End for
End if
End for

2. Return strong association rules

END.

Eg:

{Internet, streaming} = 3

{Internet} = 4

{streaming} = 4

confidence (Internet → streaming)

$= 3/4 \times 100 = 75\%$.

confidence (streaming → Internet)

$= 3/4 \times 100 = 75\%$.

For min confidence = 60%.

Internet → streaming = 75%.

streaming → Internet = 75%

cloud storage → streaming = 2/3 = 66.67%

4. Constraint-Based Frequent Itemset Mining

Input :

Transaction database D

Minimum support min_support

Required item = python

Output :

Frequent itemset containing python.

Begin :

1, Generate candidate 1-itemset.

2, Remove every candidate that does not contain python.

3, calculate support {(python)}.

4, If support {(python)} >= Min_support then

Add {python} to frequent itemset

End if

5, Set K = 2.

6. while frequent of size k-1 exist DO

Generate candidate k-itemset

For each candidate c DO

If python is not present in c then

prune c.

Else

calculate support (c).

If support (c) >= Min-support then

add c to frequent itemset.

Else

prune c.

End If

End If

End for

set k = k + 1

End while

7. Return frequent itemset contain python.

END

possible candidates;

{Python}

{Python, database}

{Python, AI}

{Python, database, AI}

5. Multilevel association Rule Mining

Input;

Transaction database D

concept hierarchy H

minimum support min_support

minimum confidence min-confidence

output;

Multilevel association rules.

Begin;

1, define the concept hierarchy H.

2, start at the highest hierarchy level.

3, Generalize transaction according to H.

4, Generate frequent itemset at the current level.

5, calculate support for each item.

7, Generate association rules

8, calculate confidence for each rule.

9, Repeat steps

10, Return association rule from all hierarchy levels

END.

support at different levels ;

Healthcare = 5/5 = 100%.

cardiology = 3/5 = 60%.

orthopedics = 2/5 = 40%.

ECG = 2/5 = 40%.